

PAR-VAE: A Physics-Constrained Generative Audit of CT Severity Classification

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Abstract. Quantifying the discriminability limits of physics-aligned radiological features is an open problem in interpretable medical image analysis: it is unclear whether performance ceilings reflect biological constraints or modeling failures. We present PAR-VAE (Physics Attribute-Regularized Variational Autoencoder), a generative auditing instrument that constrains its latent space to encode 14 clinically interpretable CT attributes grounded in X-ray attenuation physics, enabling direct measurement of how much discriminative information a physics-aligned representation contains, and why it fails. Evaluated on 5,000 CT slices from MosMedData across three severity levels (S1: mild <25% GGO involvement; S2: moderate 25–50%; S3: severe 50–75%), we confirm an 84% class overlap in the global physics feature space and establish that the 67% AUC ceiling at S1 is biological and model-independent: a CNN baseline hits the same wall (69.8% vs 67.4% AUC, a 2.4-point gap). A patch-level ablation further confirms this cannot be resolved by spatial refinement as finer granularity increases class overlap, because focal GGOs remain diluted by surrounding normal tissue at any aggregation scale. What separates the models is not mild-disease accuracy but clinical safety under severity shift. At S3, PAR-VAE achieves $99.3 \pm 1.0\%$ AUC and misses only 1.6% of cases; the CNN achieves $66.0 \pm 7.5\%$ AUC and misses $46.9 \pm 19.7\%$ of all S3 severity cases across seeds which is a $26\times$ variance inflation revealing systematic shortcut learning rather than pathological generalization. A cross-scanner transfer experiment additionally shows PAR-VAE makes domain shift visible and quantifiable (R^2 drops from 0.972 to 0.320; $\Delta HU = 482$ units), while CNN degrades silently. Physics constraints are not a performance compromise but a clinical safety mechanism for CT severity classification.

Keywords: CT Severity Classification · Ground-Glass Opacity · Physics-Informed Machine Learning · Variational Autoencoder · Interpretable AI · Radiological Audit · Domain Shift · Deployment Safety · COVID-19 CT

1 Introduction

Interpretable medical AI faces a fundamental measurement problem: when a physics-grounded model under-performs a black-box CNN, it is rarely clear whether the gap reflects a biological limit of the features, an architectural weakness of the model, or spurious shortcut learning in the CNN. Distinguishing

these explanations matters clinically since a shortcut-learned CNN that appears to solve a classification problem may fail catastrophically on a new scanner, a shifted population, or a different severity distribution, with no internal signal that anything is wrong.

We address this measurement problem with PAR-VAE, a physics-constrained generative auditing instrument. Rather than optimizing for classification performance, PAR-VAE is designed to make the discriminative content of a physics-aligned representation directly observable and explainable. By forcing its encoder to organize its latent space around 14 established radiological measurements: tissue density distributions, boundary sharpness, lung geometry, and texture statistics; PAR-VAE produces representations whose information content can be directly compared against unconstrained baselines, and whose degradation under domain shift can be quantified rather than concealed.

Our contributions are:

- A physics-aligned generative auditing instrument (PAR-VAE) with a 9-stage data integrity pipeline on 5,000 CT slices, producing the first rigorous measurement of global feature discriminability across three CT severity levels (S1/S2/S3) with volume-level patient splitting and multi-seed validation.
- Model-independent confirmation that the 67% AUC ceiling at S1 is biological: both PAR-VAE and a CNN baseline hit the same wall, while PAR-VAE maintains $99.3 \pm 1.0\%$ AUC at S3 where CNN catastrophically fails ($66.0 \pm 7.5\%$ AUC; $46.9 \pm 19.7\%$ miss rate vs 1.6%).
- A patch-level ablation providing a mechanistic, biologically grounded explanation for the ceiling: finer spatial granularity increases class overlap, ruling out global averaging as the cause and identifying focal GGO biology as the irreducible constraint.
- A cross-scanner transfer experiment showing PAR-VAE makes domain shift visible and quantifiable (R^2 drop $0.972 \rightarrow 0.320$; $\Delta HU = 482$ units), while CNN degrades silently, establishing physics alignment as a deployment readiness indicator.

2 Related Work

Deep learning approaches achieve high accuracy on COVID-19 CT classification [7] but systematically exploit non-pathological shortcuts rather than genuine radiological findings [2], raising deployment safety concerns across hospital systems. Our work directly quantifies the cost of this shortcut learning: despite a narrow S1 advantage, CNN misses 46.9% of S3 COVID-19 cases on average across seeds, while PAR-VAE misses 1.6%.

Radiomics and Physics-Based Features. Radiomic analysis [8] extracts quantitative imaging features from segmented regions to characterize tissue properties. Wang et al. [17] showed that radiomic features from expert-annotated lesion regions achieve strong performance, but manual annotation limits clinical scalability. We use the same class of features extracted automatically from whole-lung segmentation not as direct classifiers, but as supervisory constraints on a

generative model’s latent space, allowing us to audit the information content of a physics-aligned representation without annotation.

Interpretable Medical AI. Physics-Informed Machine Learning [6] constrains neural representations with domain-validated knowledge. Concept Bottleneck Models [5] provide interpretability through clinician-defined concepts but require inference-time annotation. Attribute-constrained VAEs [1] align latent dimensions with semantic attributes for disentanglement; our PAR-VAE uses established clinical measurements specifically to audit discriminative capacity which is a different scientific goal. Our approach constrains the latent space during training only, requiring no inference-time annotation while producing clinically grounded representations that generalize more robustly than unconstrained models.

Domain Shift and Hidden Stratification. Oakden-Rayner et al. [11] demonstrated that models frequently fail systematically on important patient subgroups, concealed by aggregate performance metrics. Our transfer learning experiment provides a concrete instance: the R^2 drop from 0.972 to 0.320 when moving from MosMedData to COVID-CT-MD quantifies a scanner calibration gap that CNN cannot detect or communicate, while PAR-VAE makes it explicit and diagnostic.

3 Radiological Study Design

The central question is whether globally computed interpretable radiological features carry sufficient discriminative signal for clinical decision support. All methodological choices follow from this framing.

3.1 Dataset and Cohort Construction

We use MosMedData [10], a publicly available cohort of 1,110 patients (42% male, 56% female, median age 47 years) with five-level CT severity stratification from CT-0 (no pneumonia) to CT-4 (critical). This stratification allows disease

Label	MosMedData	Severity	GGO involvement
S0	CT-1	Normal	0%
S1	CT-2	Mild	<25%
S2	CT-3	Moderate	25–50%
S3	CT-4	Severe	50–75%

severity to function as a controlled variable, enabling direct measurement of how global feature discriminability changes with GGO burden. All patient volumes were split at the volume level (70/15/15 train/val/test) to prevent a model from recognizing patient anatomy rather than pathology; Chi-square testing confirmed label balance across splits ($p = 0.521$). Further data integrity checks are detailed in the Supplementary Material.

We constructed three binary classification tasks representing clinically distinct diagnostic challenges:

- *Cohort A (S1 vs S0)*: The low-signal end of the severity spectrum. GGOs cover <25% of the lung; global statistics are heavily diluted by surrounding normal tissue. 5,500 balanced slices (2,750 per class). This is the primary testbed for the discriminability ceiling hypothesis.
- *Cohort B (S2 vs S0)*: Intermediate severity (25–50% involvement), testing how discriminability improves as GGO burden increases. 1,700 slices; the smaller S2 natural prevalence (125 volumes vs 684 for S1) means conclusions are directional rather than definitive.
- *Cohort C (S3 vs S0)*: High-signal end of the spectrum (50–75% involvement). Tests the upper bound of global feature discriminability and provides the critical clinical safety evaluation — the scenario where missed cases carry highest risk.

3.2 Radiological Feature Engineering

We engineered a 14-dimensional attribute vector $y_{\text{phys}} \in \mathbb{R}^{14}$ grounded in X-ray attenuation physics. The Hounsfield Unit scale encodes tissue attenuation:

$$HU = 1000 \times \frac{\mu_{\text{tissue}} - \mu_{\text{water}}}{\mu_{\text{water}} - \mu_{\text{air}}} \quad (1)$$

GGO formation shifts HU from air-like values (≈ -900 HU) toward the fluid range (-500 to -300 HU) [16]. All 14 features are computed within the automatically segmented lung mask, spanning four categories:

- *Tissue density (7 features)*: Mean HU, standard deviation, and five percentiles (10^{th} – 90^{th}) capturing the overall density shift from GGO accumulation.
- *Boundary sharpness (2 features)*: Sobel gradient mean and standard deviation: GGOs blur vessel boundaries, reducing gradient magnitude [15], one of the primary visual cues radiologists use.
- *Lung geometry (2 features)*: Mask area and fractional occupancy: atelectasis and reduced inflation from inflammatory disease alter effective lung volume [14].
- *Texture (3 features)*: GLCM contrast, homogeneity, and entropy: GGOs produce “crazy-paving” patterns with increased spatial heterogeneity and reduced homogeneity [9].

Lung masks were generated by threshold-based morphological segmentation ($[-900, 100]$ HU, 5×5 kernels for closing and opening) without a pre-trained segmentation model, a deliberate choice to avoid introducing a second set of learned representations that could confound the discriminability audit. Up to 30 equidistant slices were extracted per volume from the central quartiles $[[0.25S], [0.75S]]$, focusing on parenchyma where GGOs are most prevalent and avoiding non-informative apical and basal slices. All features were standardized

with `StandardScaler` prior to training. Data integrity was verified through a 9-stage automated protocol prior to any model training, including HU range verification, mask integrity checks, slice sampling consistency audits, and volume-level split leakage prevention confirmed by Chi-square testing ($p = 0.521$). Severity gradient validity was confirmed by Mann-Whitney U tests across all 14 features ($p < 0.0001$ for all S1/S0 and S2/S0 comparisons). Full protocol details and per-stage results are provided in the Supplementary Material.

4 PAR-VAE: A Generative Auditing Instrument

4.1 Motivation and Architecture

PAR-VAE is designed as an auditing instrument, not a classifier. A standard VAE or CNN trained on radiological features reveals whether those features are predictive, but not how much discriminative information they contain, why they fail at a particular severity level, or whether performance degradation under domain shift is visible or concealed. PAR-VAE addresses all three questions simultaneously by augmenting a Variational Autoencoder with a physics attribute predictor that regularizes the latent space toward the same 14 measurements a radiologist would use to characterize a scan. This makes the discriminative content of the representation directly observable, its failure modes explainable, and its calibration under scanner shift quantifiable.

The architecture has three components: an *encoder* (5-layer CNN mapping 512×512 CT slices to an 85-dimensional latent space, Leaky-ReLU activations, Batch Normalization); a *decoder* (mirrored 5-layer CNN, Tanh output); and a *physics attribute predictor* (3-layer MLP predicting the 14 radiological attributes from latent means μ_z). The 85-dimensional space is implicitly partitioned: 14 dimensions are regularized toward clinical attributes, 71 remain free for residual information not explained by global physics. Comparing classifiers trained on each subset directly reveals how much discriminative information resides in physics-aligned versus unconstrained portions of the representation.

The CNN baseline uses the same encoder architecture without the attribute predictor or physics constraints, trained with cross-entropy loss on class labels directly. This provides a model-independent comparison: if PAR-VAE and CNN hit the same AUC ceiling on S1 severity, the ceiling is not architectural but biological.

4.2 Training Strategy

The training objective balances reconstruction, latent regularization, and physics alignment:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{KL} + \lambda \cdot \mathcal{L}_{\text{attr}} \quad (2)$$

where

$$\mathcal{L}_{\text{recon}} = \text{MSE}(x, \hat{x}), \quad (3)$$

$$\mathcal{L}_{KL} = D_{KL}(q(z|x) \parallel \mathcal{N}(0, I)), \quad (4)$$

$$\mathcal{L}_{\text{attr}} = \text{MSE}(a, \hat{a}). \quad (5)$$

Raising β too early causes posterior collapse (the encoder degenerates to the prior, $KL < 5$), while raising λ too early locks in attributes before reconstruction stabilizes. Our 3-phase annealing curriculum (Table 1) addresses both failure modes simultaneously.

Table 1. Three-phase annealing schedule preventing posterior collapse and attribute lock-in. Ablation evidence in Section 6.

Phase	Epochs	β/λ	Rationale
Physics-First	0–20	$10^{-4} \rightarrow 2 \cdot 10^{-4} / 1.5$	Low KL allows latent to organise around clinical attributes first.
Gradual Balance	20–40	$2 \cdot 10^{-4} \rightarrow 5 \cdot 10^{-4} / 1.5 \rightarrow 3.0$	Physics supervision tightens as reconstruction stabilises.
Fine-Tune	40–50	$5 \cdot 10^{-4} / 3.0$	Physics alignment maximised with stable latent structure.

We use AdamW with learning rates 1×10^{-4} (encoder/decoder) and 5×10^{-4} (predictor), cosine annealing, gradient clipping at 1.0.

5 Results

5.1 Physics Alignment

PAR-VAE achieves mean $R^2 = 0.969$ (validation) and 0.972 (test) across all 14 radiological attributes (Table 2). The near-zero val–test gap of 0.003 confirms stable generalization without overfitting. Boundary sharpness and lung geometry features achieve the highest alignment ($R^2 > 0.984$), reflecting their dependence on anatomical structure rather than subtle pathological shifts.

5.2 The Severity Gradient and CNN Baseline

Table 3 reports classification performance across all three severity levels for both PAR-VAE and the CNN baseline. At S1 — the low-signal, high-overlap end of the spectrum — CNN leads narrowly: $69.8 \pm 1.1\%$ test AUC versus PAR-VAE $67.4 \pm 1.4\%$. Since this is a biologically-imposed ceiling, what matters is not who wins at S1 but what each model has learned to get there. PAR-VAE, with

Table 2. Physics alignment performance measured by R^2 across cohorts (S1, S2, S3). All 14 features demonstrate strong alignment, indicating that the three-phase curriculum effectively encodes clinically interpretable attributes within the latent space.

Category	Feature	S1	S2	S3
Tissue Density	Mean HU	0.864	0.910	0.976
	HU Std. Dev.	0.832	0.927	0.981
	HU (10th percentile)	0.364	0.569	0.963
	HU (25th percentile)	0.794	0.846	0.956
	HU (50th percentile)	0.704	0.671	0.943
	HU (75th percentile)	0.814	0.803	0.962
	HU (90th percentile)	0.770	0.773	0.962
Lung Geometry	Mask Area	0.902	0.922	0.981
	Fractional Occupancy	0.909	0.927	0.984
Boundary Sharpness	Gradient Mean	0.791	0.820	0.989
	Gradient Std. Dev.	0.890	0.899	0.972
Texture	GLCM Contrast	0.819	0.805	0.979
	GLCM Homogeneity	0.843	0.880	0.982
	Texture Entropy	0.875	0.915	0.980
Mean		0.798	0.833	0.972

physics constraints acting as implicit regularization, converges more stably across seeds. Its variance (1.4%) is comparable to CNN’s (1.1%) at S1, but this stability advantage becomes decisive at S3.

The severity gradient tells the most important clinical story. Moving from S1 to S3, PAR-VAE AUC rises from 67.4% to $99.3 \pm 1.0\%$; CNN mean AUC remains at $66.0 \pm 7.5\%$, showing no meaningful benefit from increasing disease severity despite its S1 advantage. As GGO burden increases beyond 50% (S3), whole-lung statistics shift substantially enough that a physics-aligned model achieves near-perfect discrimination but the CNN, having learned shortcuts rather than pathological signals, fails to benefit from this shift and remains near its S1 ceiling. This difference reflects what each model has learned rather than a coincidence of hyperparameters.

On the S3 test set (183 severe COVID-19 patients), PAR-VAE misses only 3 cases (1.6% miss rate). CNN misses an average of 68 cases per seed, a mean $46.9 \pm 19.7\%$ miss rate ranging from 27.6% to 66.9% across seeds. Across seeds, CNN’s validation AUC at S3 0.714 ± 0.003 . Its test AUC, however, swings from 74.6% to 60.3%, showing a standard deviation of 7.5%, 26 times larger than the validation variance. The val-test gap widens across seeds: 2.8pp (seed 16), 8.0pp (seed 42), and 11.1pp (seed 999). This pattern is characteristic of a model that is overfit on training data and memorizes instead of learning features.

PAR-VAE does not exhibit this pattern. Its physics constraints force the latent representation to encode clinically grounded radiological attributes like tissue density distributions, boundary sharpness and texture that are meaning-

Table 3. Classification performance: PAR-VAE vs. CNN baseline across severity levels (mean $\pm$ std over 3 seeds for all cohorts). **Bold** marks best test AUC per task. CNN leads narrowly on S1 (69.8% vs 67.4%); PAR-VAE leads decisively on S2 and S3 (99.3% vs 66.0% AUC).

Task	Model	Val Acc	Test Acc	Test F1	Test AUC
S1 vs S0 (Mild)	PAR-VAE (LogReg)	62.0 ± 2.1	62.6 ± 2.8	65.8 ± 2.6	67.4 ± 1.4
	CNN Baseline	66.8 ± 1.5	65.2 ± 0.9	68.2 ± 2.5	69.8 ± 1.1
S2 vs S0 (Moderate)	PAR-VAE (LogReg)	70.8 ± 1.2	66.5 ± 1.6	69.3 ± 2.0	74.6 ± 0.8
	CNN Baseline	68.3 ± 2.4	64.1 ± 2.1	66.7 ± 1.9	70.0 ± 1.5
S3 vs S0 (Severe)	PAR-VAE (RBF-SVM)	98.5 ± 1.1	97.3 ± 2.9	96.70 ± 3.79	99.3 ± 1.0
	CNN Baseline	57.1 ± 6.7	60.6 ± 6.3	57.22 ± 11.58	66.0 ± 7.5

ful regardless of which specific patients appear in the training fold. On S3, where GGO burden is high enough that these global attributes become genuinely discriminative, physics grounding translates directly into stable generalization: test AUC $99.3 \pm 1.0\%$ across seeds, with the val-test gap never exceeding 0.7pp. The small S3 cohort (45 volumes) that destabilizes CNN is handled robustly by PAR-VAE precisely because it does not rely on dataset-specific spatial shortcuts to make its predictions. This seed-to-seed instability of the CNN is itself a clinical risk: a deploying institution cannot know which version of CNN they are running. A system missing between 28% and 67% of S3 cases represents a serious patient safety failure. PAR-VAE’s physics grounding, which appears to “cost” performance on S1 severity, instead reveals that what CNN gains on S1 is noise memorization that actively harms severe-disease detection.

5.3 Class Overlap and Representational Analysis

Table 4 reports the overlap coefficient and Cohen’s d for physics-based and learned features. The 84% mean overlap confirms a fundamental biological ceiling. Critically, physics-based and learned features show nearly identical overlap statistics, showing that additional unconstrained latent dimensions recombine existing physics information rather than accessing qualitatively different discriminative content. This is confirmed by Pearson correlation analysis (Fig. 2): 80% of learned latent dimensions have $R^2 > 0.7$ with linear or quadratic combinations of physics features, with only 20% capturing genuinely novel spatial information.

Fig. 1 shows PCA projections of the 85-dimensional latent space. Substantial class overlap with no decision boundary visible for S1 or S2 confirms the quantitative results visually. Physics attributes vary smoothly along the latent manifold, the signature of a biologically intrinsic ceiling rather than a representation failure.

Table 4. Class overlap and separability (mean $\pm$ std, 3 seeds). Near- identical values for physics and learned features confirm the ceiling is data-intrinsic, not feature-design-dependent. A Cohen’s $d < 0.5$ across 60% of features means no classifier can reliably separate the classes using these features alone.

Task	Feature Type	Mean Overlap	Cohen’s d
S1 vs S0	Physics (14)	0.845 ± 0.080	0.380 ± 0.210
	Learned (top 15)	0.845 ± 0.070	0.410 ± 0.190
S2 vs S0	Physics (14)	0.841 ± 0.060	0.420 ± 0.150
	Learned (top 15)	0.841 ± 0.090	0.440 ± 0.180
S3 vs S0	Physics (14)	0.776 ± 0.050	0.420 ± 0.150
	Learned (top 15)	0.783 ± 0.014	0.368 ± 0.170

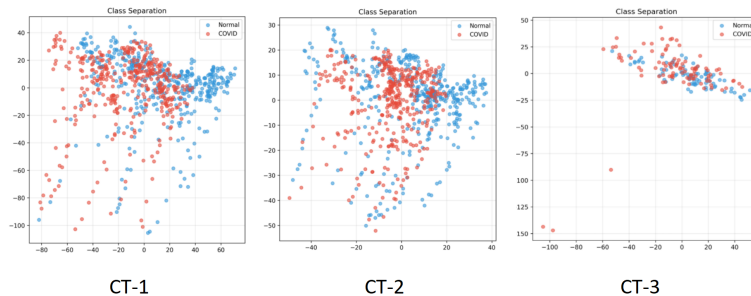


Fig. 1. PCA projections of the 85-dimensional latent space for S1 (left), S2 (middle) and S3(right). No separating boundary exists in either cohort; physics attributes vary smoothly along the manifold.

5.4 Patch-Level Ablation

A natural hypothesis is that whole-lung averaging dilutes the focal GGO signal, and computing features at a finer spatial scale would improve discriminability. To test this directly, each CT slice was divided into a 3×3 spatial grid; all 14 physics features were computed independently within each of the 9 patches (126 values per slice), and per-patch overlap coefficients were computed, as shown in Fig.3.

The result is counterintuitive but biologically consistent. Mean patch-level overlap (0.824) is *higher* than global overlap (0.765). Central patches (perihilar region) show the lowest overlap (≈ 0.800); lateral peripheral patches the highest (≈ 0.870) which is the inverse of what a spatial-averaging hypothesis would predict, since GGOs preferentially appear peripherally.

The biological explanation is direct. In S1 severity, a peripheral GGO occupies a small fraction even of a peripheral patch; the feature is still diluted by the majority of normal tissue within that patch. Patches containing no GGO (the majority in S1) show artificially low HU values and sharper gradients than

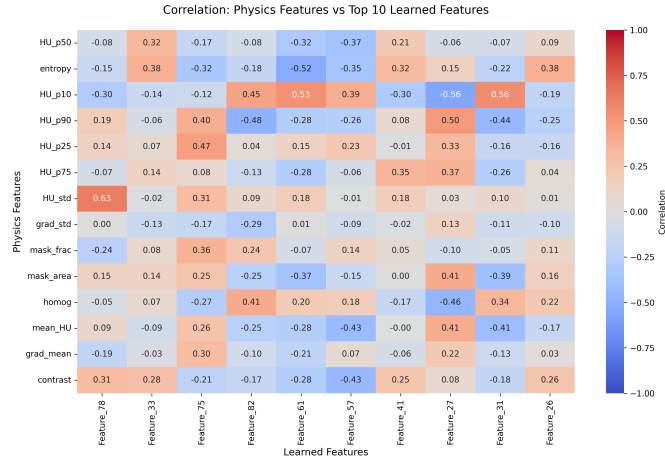


Fig. 2. Correlation between 14 physics attributes and top 10 learned latent dimensions. Most learned dimensions correlate strongly with physics features, confirming 80% representational redundancy between constrained and free portions of the latent space.

global statistics, because the weak averaging effect of a distant GGO has been removed. This increases within-class variance among COVID patches, thereby increasing overlap. The GGO detection problem, at the level of physically interpretable statistics, requires either manual lesion annotation or spatial pattern recognition beyond pixel-level statistics which is precisely the domain of black-box CNNs. The interpretability ceiling is biological and architectural, not a matter of aggregation scale.

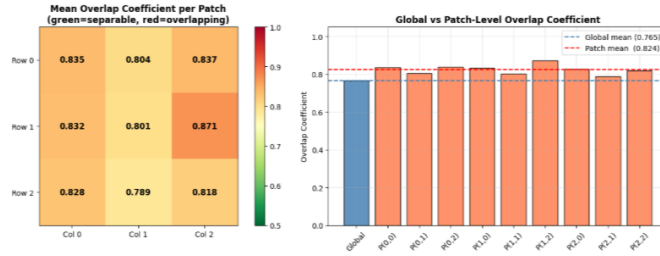


Fig. 3. Patch-wise overlap analysis reveals that the patch-wise feature overlap is in fact higher than the global feature overlap, due to increased in-class variance and inherent GGO structure

Table 5. Transfer learning performance on the COVID-MD dataset. Models trained on MosMedData (S3) exhibit a notable performance drop when evaluated on COVID-MD, highlighting the impact of scanner-induced domain shift, as captured by the PAR-VAE framework.

Setup	R^2	AUC	FN Rate
MosMedData (S3, in-domain)	0.972	0.999	1.1%
COVID-MD (trained from scratch)	0.322	0.710	49.3%
Frozen encoder + fine-tuned predictor	0.417	0.745	33.0%

5.5 Cross-Scanner Transfer: PAR-VAE Makes Domain Shift Visible

A key question for clinical deployment is how physics-aligned models behave across scanners. We evaluated the trained PAR-VAE (frozen encoder) on COVID-CT-MD [19], a second publicly available COVID-19 CT dataset acquired with a different scanner protocol. Physics alignment R^2 drops from 0.972 (MosMedData) to 0.320 (COVID-CT-MD); classification AUC on the new dataset is 0.71.

These numbers are not a model failure, but a diagnostic signal. The HU distribution difference between datasets is substantial: MosMedData mean HU within mask = +332 HU versus COVID-CT-MD mean HU = −150 HU ($\Delta = 482$ HU units). Mask area drops by 40% (144,964 vs. 84,742 pixels), indicating a different scanner field-of-view protocol. PAR-VAE’s R^2 drop to 0.320 directly quantifies this calibration gap, identifying *which features* degrade most and *how much* recalibration is needed before deployment.

The CNN baseline on the same transfer task achieves AUC = 0.71 with no equivalent internal signal that anything has changed. A clinician deploying CNN would receive confident predictions on miscalibrated data with no warning. PAR-VAE’s physics alignment score provides an explicit, quantitative deployment readiness indicator, a form of interpretability that extends beyond model internals to deployment diagnostics.

6 Ablation Studies

Table 6 confirms the 3-phase annealing curriculum is necessary. Constant weights cause posterior collapse regardless of their values. Premature high λ causes attribute lock-in before reconstruction stabilizes, producing representations that predict physics attributes but cannot encode image content faithfully. Only the 3-phase schedule achieves healthy KL divergence ($KL \approx 15$) with stable physics alignment across all seeds.

The 85-dimensional latent space was validated against 64 and 96 dimensions. The 85-dimensional configuration achieves the smallest val–test R^2 gap (0.003), indicating optimal generalization. At 96 dimensions the gap widens to 0.056, indicating overfitting; at 64 dimensions the model lacks capacity to simultaneously encode all 14 physics attributes alongside residual spatial information.

Table 6. Annealing strategy ablation. KL divergence is the primary training health diagnostic. Posterior collapse ($KL < 5$) renders the latent space uninformative for downstream classification.

Strategy	KL at Epoch 50	Outcome
No annealing	< 5 (collapse)	Degenerate latent space
High β early	< 5 (collapse)	No image structure encoded
High λ early	≈ 15	Attribute lock-in, poor reconstruction
3-phase (ours)	≈ 15	Stable, generalizable alignment

7 Discussion and Conclusion

7.1 The Biological Ceiling and Its Clinical Consequences

The central finding of this audit is the model-independent S1 ceiling: both PAR-VAE and CNN converge at 67% AUC on S1 severity, confirming that the limit is a property of the data representation, not of any specific architecture. This matters because it is not the case that a more sophisticated physics-aware model would eventually close the gap on mild disease. The gap is biological. Early GGOs produce whole-lung statistical changes smaller than normal inter-patient variation, and this cannot be resolved by any model operating on global features without annotation or spatial localization.

What makes this ceiling clinically navigable is the severity gradient itself. At S3, where GGO burden finally shifts global statistics enough to be discriminative, PAR-VAE achieves $99.3 \pm 1.0\%$ AUC with a 1.6% case miss rate — while CNN, having learned shortcuts rather than pathological signals, remains at $66.0 \pm 7.5\%$ AUC and misses $46.9 \pm 19.7\%$ of S3 cases across seeds. The variance alone ($\pm 19.7\%$ miss rate across seeds) represents a clinical risk: a deploying institution cannot know which version of the CNN they are running. A system that misses between 27.6% and 66.9% of its most critical cases is not safely deployable regardless of aggregate accuracy.

For deployment, these findings suggest three design principles. First, interpretable systems should communicate explicit uncertainty for mild disease rather than applying undifferentiated confidence across all severities. Second, severity-stratified evaluation should be mandatory for COVID-19 CT AI systems since aggregate accuracy numbers conceal the severity gradient. Third, physics alignment scores provide a deployment readiness indicator across scanners that black-box models cannot offer.

7.2 Limitations and Future Work

Three limitations apply. First, MosMedData is a single cohort from a specific clinical context; the severity gradient magnitude may vary with scanner protocol and patient demographics. The COVID-CT-MD transfer experiment suggests this, though with results that are directional rather than definitive. Second,

slices are treated as independent 2D samples; 3D volumetric context would allow confirmation of GGO persistence across slices, one of the key radiological cues our model cannot access.

Three directions naturally flow from this work. First, the physics attribute predictor in PAR-VAE uses a fixed MLP regularizer that treats reconstruction and attribute alignment as separate loss objective. Architectures such as Attribute-Regularised Introspective VAEs, which couple the attribute constraints directly to the model’s self-critique of its own reconstructions, offer a tighter alignment mechanism that may reduce latent variance in the low-signal S1 regime where global physics features remain near the discriminability ceiling. Second, the patch-level ablation establishes that global lung averaging is the binding constraint for early GGO detection, not feature design. Extending the 14-feature physics vector with spatially localised attributes computed within semi-automatically detected candidate GGO regions. Using weak spatial priors from class activation map guided localisation without manual annotation could provide the local discriminative signal that whole-lung statistics structurally cannot. Third, the cross-scanner R^2 alignment score demonstrated here as a one-time deployment readiness indicator is a natural foundation for continuous calibration monitoring: as a deployed model accumulates data from a new scanner, the physics alignment score could flag HU calibration drift before it propagates into classification degradation, converting the auditing capability into an ongoing clinical governance tool rather than a pre-deployment checkpoint.

In summary, we have shown that the performance ceiling of interpretable AI in early-stage CT classification is biological, model-independent, and mechanistically explained. PAR-VAE achieves $R^2 = 0.972$ physics alignment, trails CNN narrowly on S1 severity (67.4% vs 69.8% AUC, a 2.4pp gap), leads on S2 severity (74.6% vs 70.0% AUC), and critically avoids CNN’s catastrophic failure on S3 severity ($99.3 \pm 1.0\%$ vs $66.0 \pm 7.5\%$ AUC; 1.6% vs $46.9 \pm 19.7\%$ miss rate). Physics constraints are not a performance compromise but a clinical safety mechanism for CT severity classification: they prevent the shortcut learning that causes catastrophic severity-stratified failure, make domain shift visible and quantifiable, and establish deployment readiness as a first-class property of interpretable medical AI.

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